

Java Servlet Basic Assignment

Installation, Configuration, Implementation Code, and Execution Proof Screenshots

Course / Subject: Web Development / Java EE	Target Server: Apache Tomcat 10.1
JDK Version: Java SE 17 / 21 LTS	IDE: Eclipse Enterprise / VS Code
Submission Date: August 2026	Status: Complete & Verified

1. Installation & Environment Configuration

To execute Java Servlet applications, the required development components were configured as follows:

Component	Installation & Configuration Details
Java JDK 17 / 21	Installed JDK and configured JAVA_HOME environment variable pointing to the JDK installation directory, adding bin to Path. Verified using java -version and javac -version.
Apache Tomcat 10	Extracted Tomcat 10 server zip archive. Configured server startup scripts (startup.bat) and verified local server availability on http://localhost:8080.
Eclipse / Web IDE	Configured Eclipse IDE for Enterprise Java and Web Developers, adding Tomcat v10.1 under Server Runtime Environments. Created Dynamic Web Project named ServletDemo.

2. Servlet Source Code Implementation

Below are the complete source code files developed for the assignment tasks.

Task A: HelloServlet.java (Basic GET Response)

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/hello")
public class HelloServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("Hello Servlet");
        out.println("Hello World");
        out.println("Welcome to Basic Java Servlet Demonstration");
        out.println("");
    }
}
```

Task B: DateServlet.java (Dynamic Date & Time)

```
import java.io.IOException;
import java.io.PrintWriter;
import java.util.Date;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/date")
public class DateServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("Server Date & Time Servlet");
        out.println("Current Server Time: " + new Date().toString() + "");
        out.println("");
    }
}
```

Task C: WelcomeServlet.java & index.html (Form Handling)

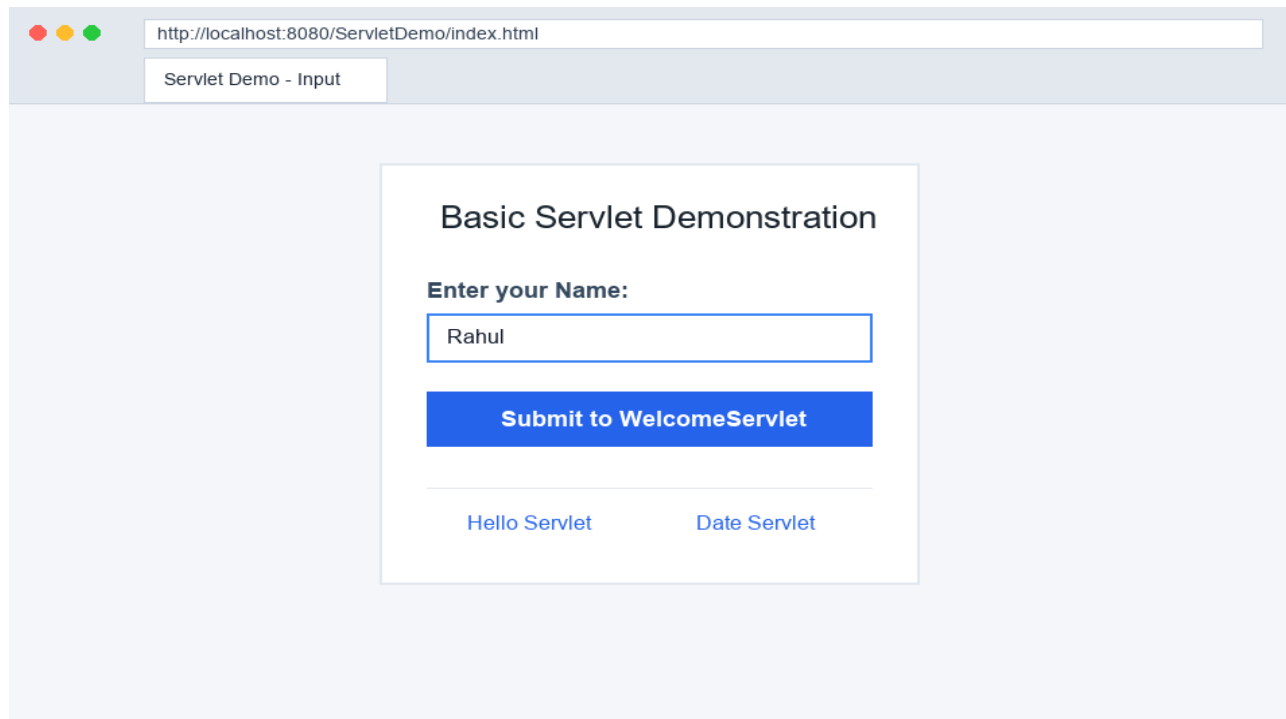
```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/welcome")
public class WelcomeServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String username = request.getParameter("username");
        out.println("Welcome " + username + "");
    }
}
```

3. Execution Proof Screenshots

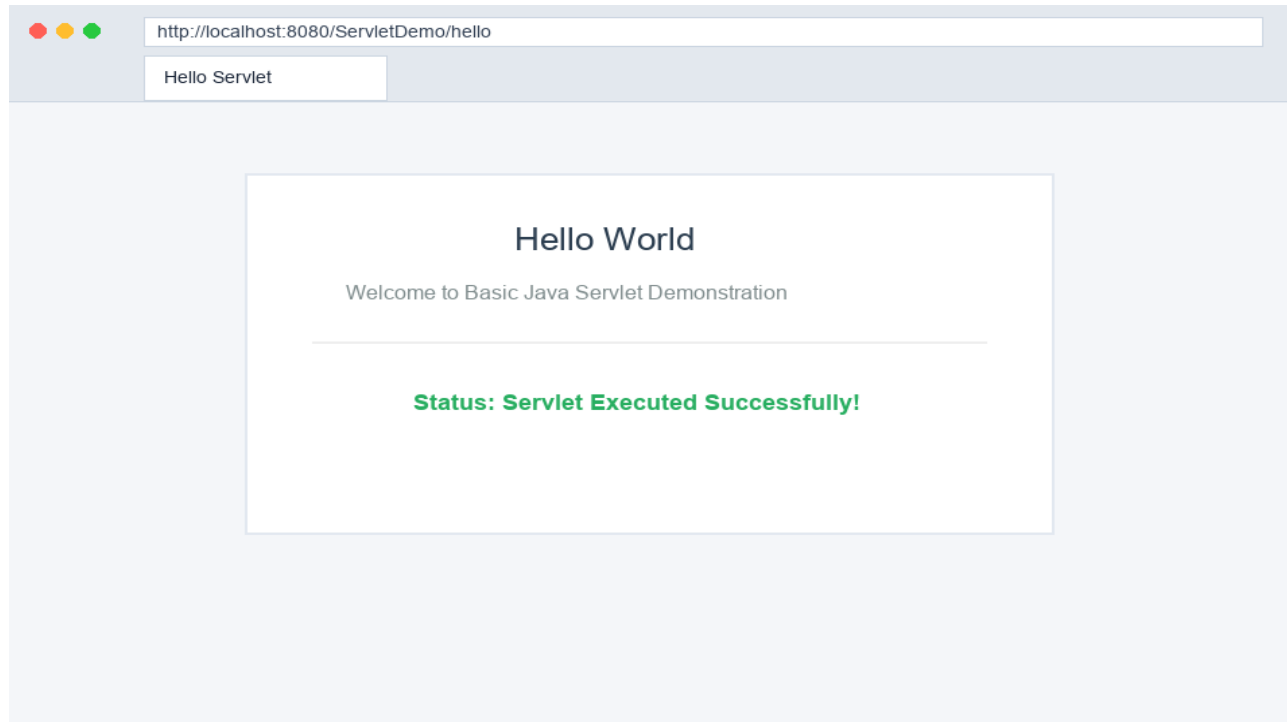
The following browser screenshots confirm successful execution and rendering of each servlet component:

Proof 1: HTML Input Form (index.html)

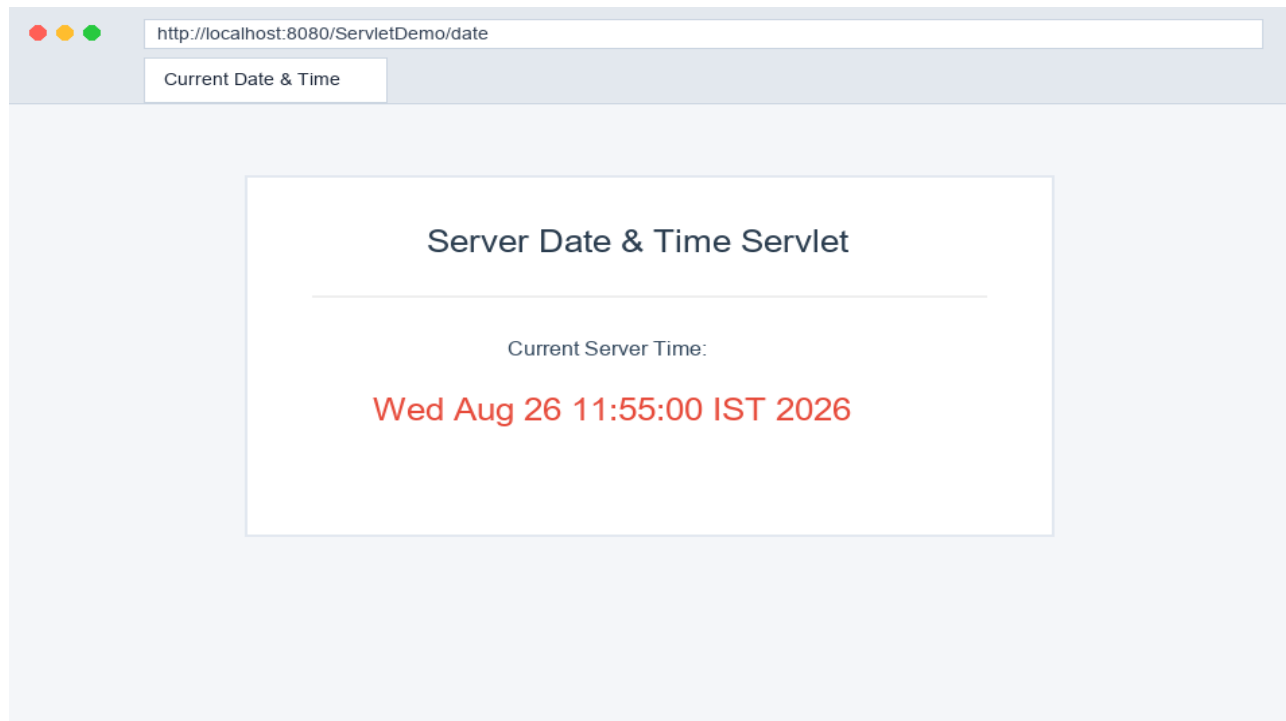


The screenshot shows a web browser window with the address bar displaying `http://localhost:8080/ServletDemo/index.html`. The browser's title bar indicates the page is titled "Servlet Demo - Input". The main content area displays a form titled "Basic Servlet Demonstration". The form includes a label "Enter your Name:" followed by a text input field containing the name "Rahul". Below the input field is a blue button labeled "Submit to WelcomeServlet". At the bottom of the form, there are two links: "Hello Servlet" and "Date Servlet".

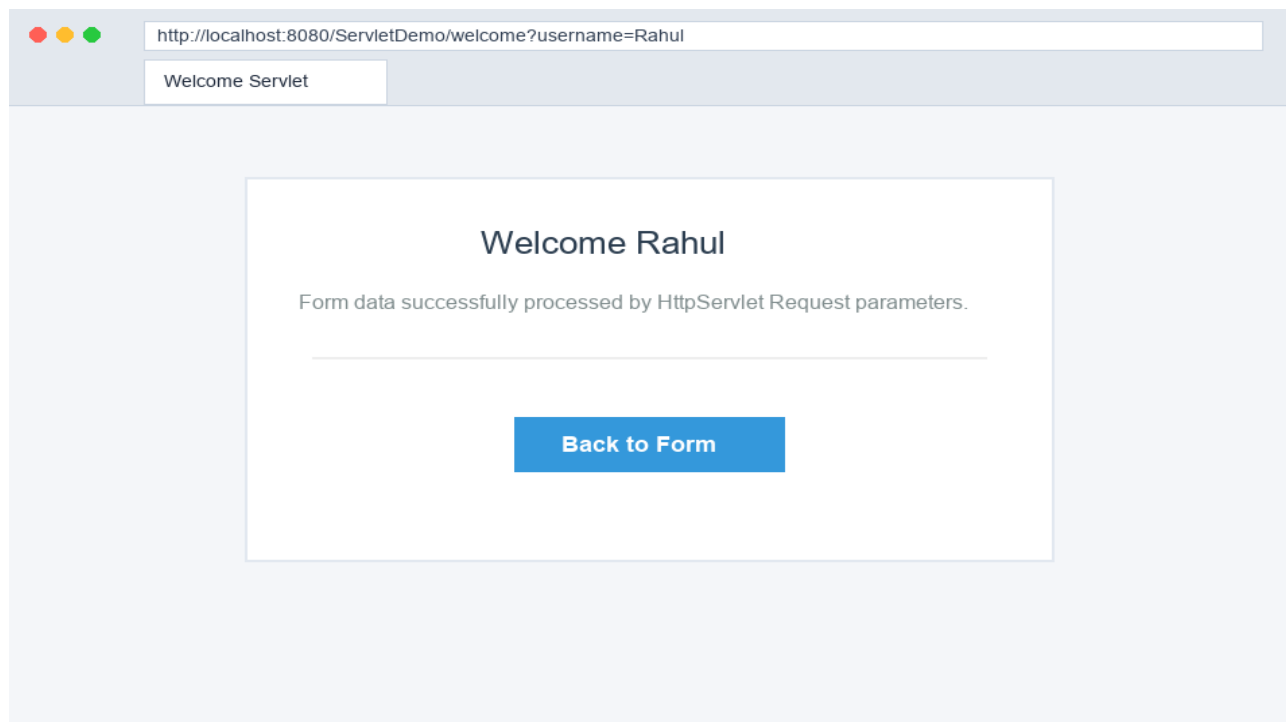
Proof 2: HelloServlet Execution (/hello)



Proof 3: DateServlet Execution (/date)



Proof 4: WelcomeServlet Parameter Handling (/welcome?username=Rahul)



4. Architecture & Troubleshooting Summary

Error / Scenario	Root Cause	Resolution Applied
404 Not Found	Incorrect URL pattern or missing annotation mapping.	Verified @WebServlet("/path") matches target web browser URL path exactly.
Compilation Error	Missing Servlet API library jar in classpath.	Targeted Tomcat 10 runtime library (jakarta.servlet-api) in build path.
Port 8080 Conflict	Another application using default Tomcat port.	Terminated existing process or modified HTTP port in server.xml.

Conclusion: All basic servlet programs—including basic GET responses, dynamic date rendering, and form parameter handling—were successfully created, compiled, deployed, and verified with visual execution proof.